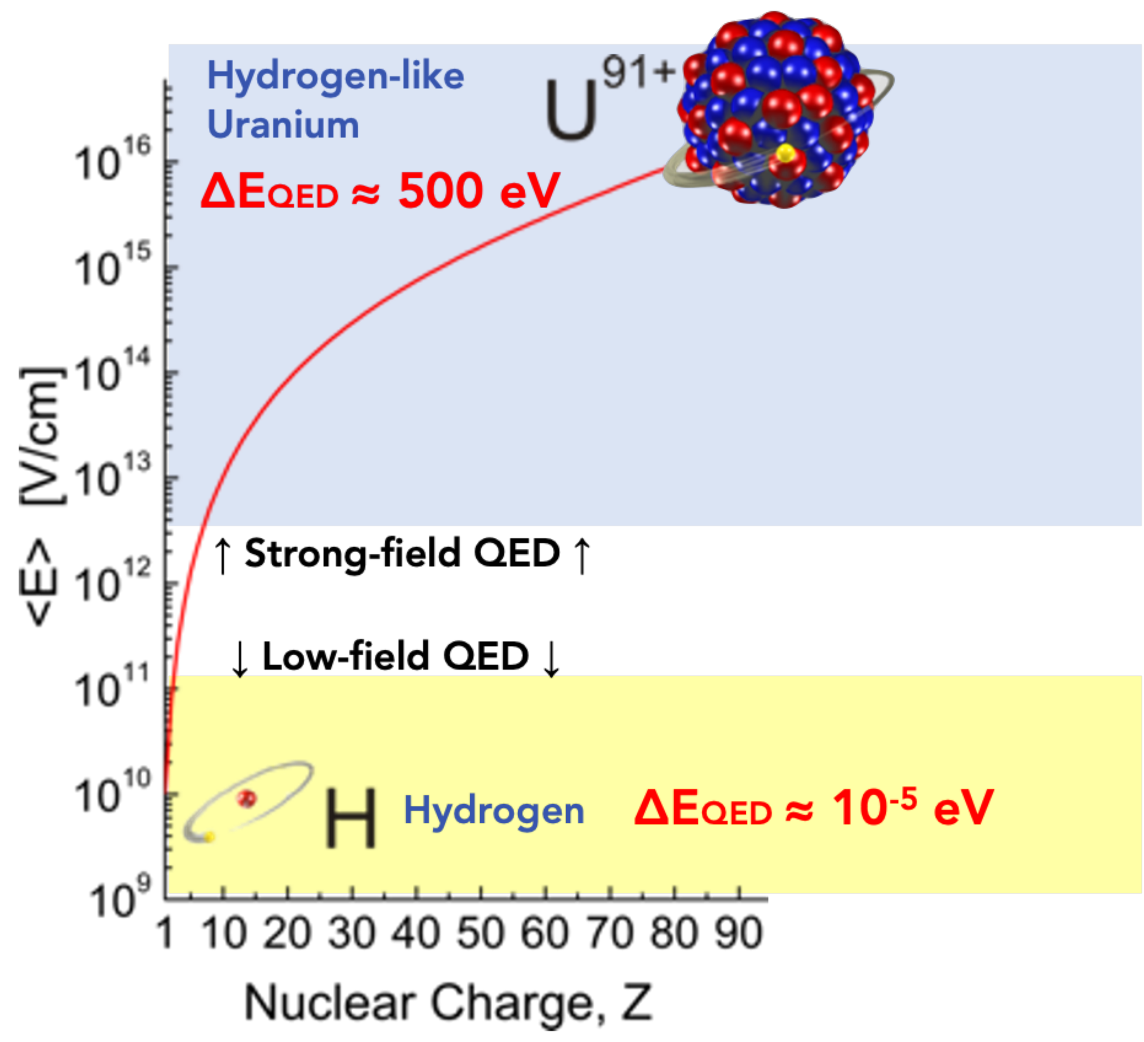
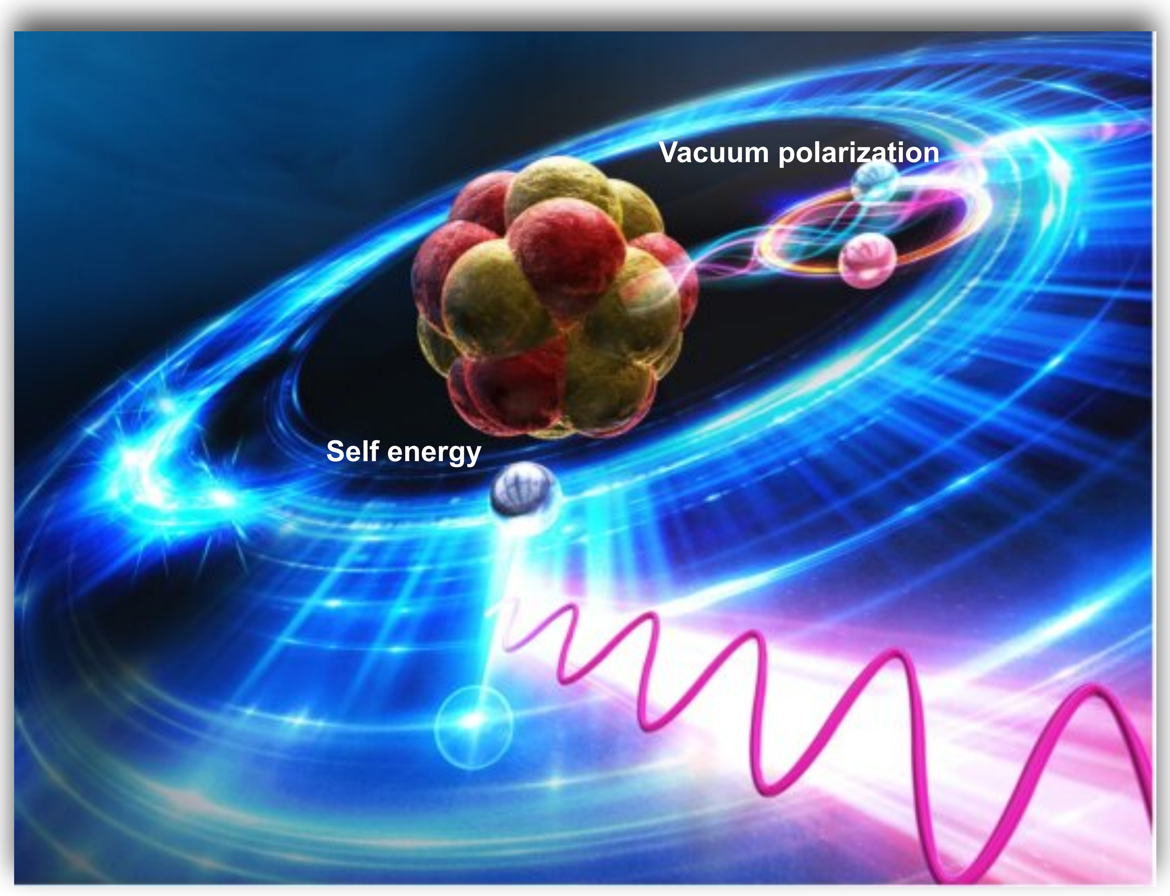
antiProtonic Atom X-ray spectroscopy

Strong field QED with antiprotonic atoms and quantum sensors

Strong-field Quantum Electrodynamics





Key physics questions:

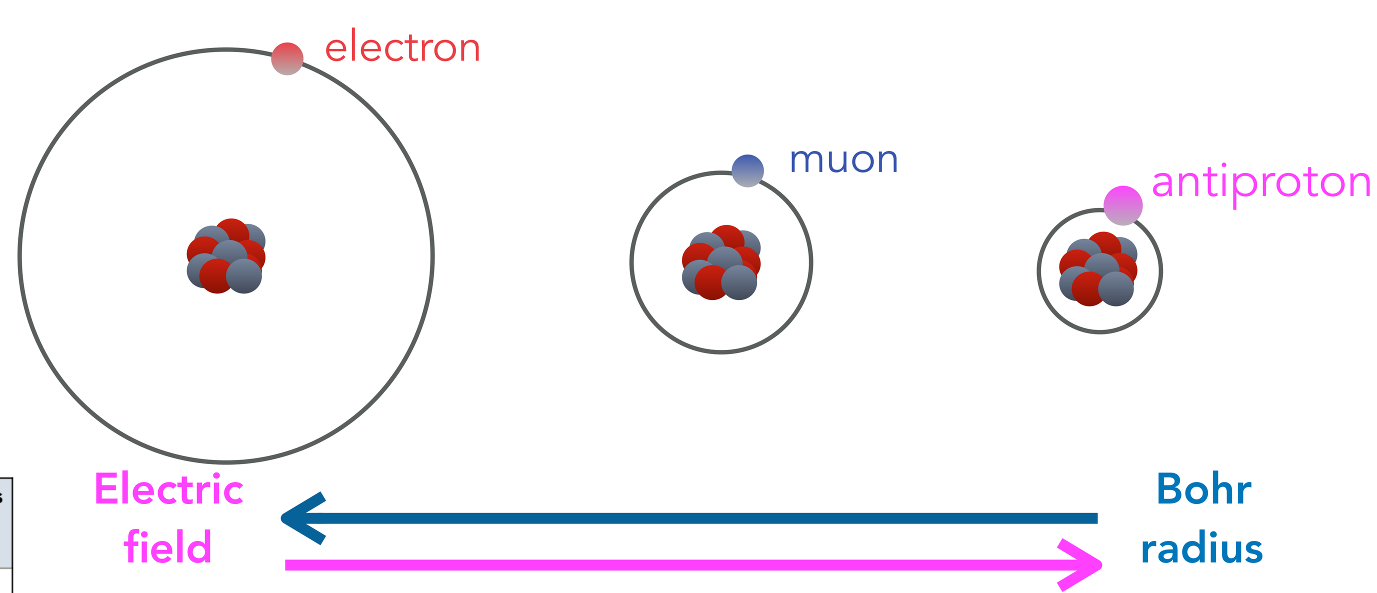
- What happens in the regime of strong-field QED for atomic systems ?
- What is the structure of exotic atoms and how are they influenced by the quantum vacuum ?
- Can we use exotic systems to look for new physics ?

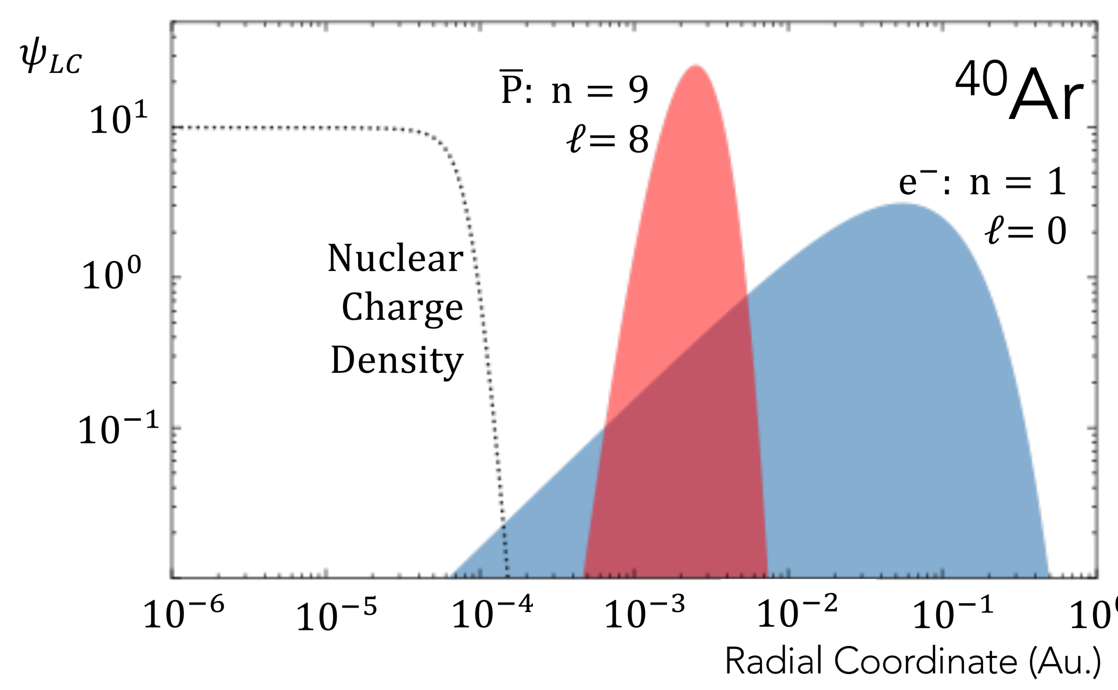
The PAX concept: Strongest field QED → highest sensitivity

- **Heavy exotic particle**
  - →Small Bohr radius
  - →Strong electric field
- **Higher order QED effects magnified**
- **Transitions between Rydberg states**
  - → Nuclear overlap minimized
  - → Pure strong-field QED

Atom	Transition	Transition energy	1st order QED	2nd order QED	Nuclear effects
H-like U	Lyman α1	~100 keV	3x10-3	1x10-5	2x10-3
antiprotonic-Xe	n=12→n=11	~100 keV	7x10-3	6x10-5	1x10-5

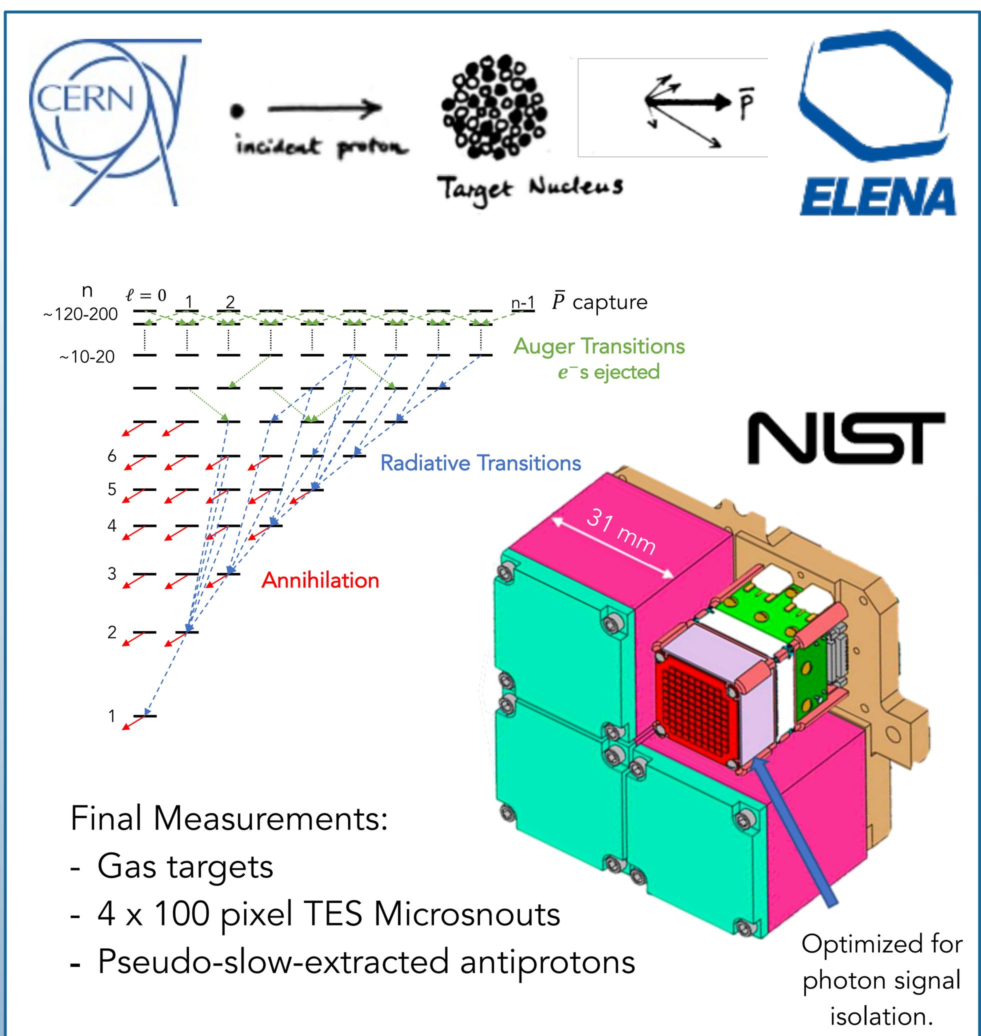
QED x 3-6 Nuclear effects / 100



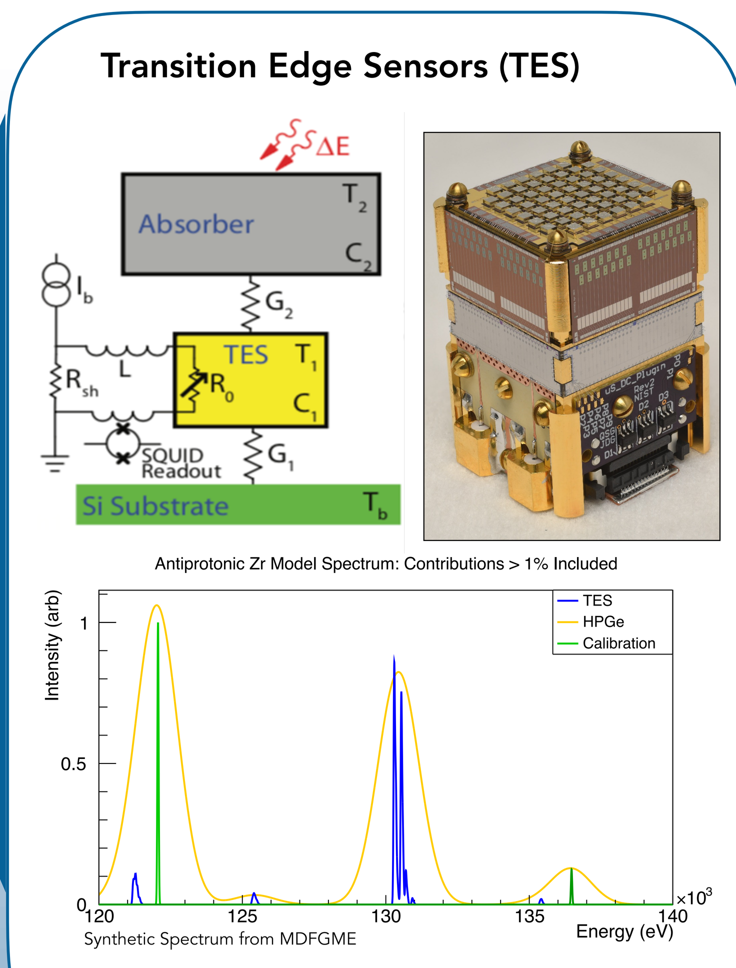


$$m_{\bar{p}} \approx 2000 m_{e^-} \implies r_{\bar{p}} \approx \frac{1}{2000} r_{e^-}$$

The PAX experimental approach : quantum sensing x-ray detectors and exotic beams



Final Measurements:
- Gas targets
- 4 x 100 pixel TES Microsnouts
- Pseudo-slow-extracted antiprotons



Latest generation TESs achieve
50 eV FWHM at 100 keV

First demonstration of high precision x-ray spectroscopy of antiprotonic atoms with quantum sensing x-ray microcalorimeter

Current line energies are consistent with predictions.

$\frac{\delta E}{E}$ in $9\ell_{\frac{17}{2}} \rightarrow 8k_{\frac{15}{2}}$ of ^{90}Zr

Stat. Unc.	$\mathbf{4.5\times10^{-5}}$
1-Loop QED	5.1×10^{-3}
2-Loop QED	$\mathbf{3.6\times10^{-5}}$
FNS	1.2×10^{-5}
10 e^- Shift	1.1×10^{-5}

Preliminary

P. Indelicato, and J. Sommerfeldt, Private Communication 2026

Preliminary results have been removed.

On target for 2-loop QED tests!

Transition ($n_i \rightarrow n_f$)	Appx. Transition energy (keV)	1st order QED	2nd order QED	Nuclear effects
^{20}Ne (6→5)	30	4 E-3	3 E-5	2 E-6
^{40}Ar (6→5)	100	5 E-3	5 E-5	1 E-5
^{84}Kr (9→8)	100	5 E-3	5 E-5	1 E-5
^{132}Xe (10→9)	170	5 E-3	5 E-5	2 E-5
^{184}W (12→11)	180	5 E-3	5 E-5	2 E-5

Highest field system ever accessed in the laboratory !

PAX firsts:

- Study second-order QED effects across 10 ≤ Z ≤ 74
- Achieve 10^{-5} experimental precision for heavy exotic atom spectroscopy

Prospectives : Strong interaction studies, exotic physics searches

With many thanks to our collaborators!

NIST, TECHNION Israel Institute of Technology, Imperial College London, RIKEN, CERN, Institute of Science Tokyo, LPNHE Paris, NOVA

Wanna learn more ? Check out :

G. Baptista et al, PoS EXA-LEAP2024, 085 (2025)
N. Paul et al. Physical Review Letters 126, 173001 (2021)
T. Okumura et al, Physical Review Letters 130, 173001 (2023)